SCIENCE · CELLS AND ENERGY

From Plate to Power

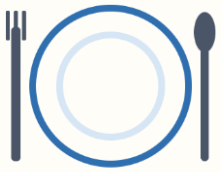
How cells get nutrients from food and turn them into energy to keep plants and animals alive

Plants | Animals | Energy



WARM-UP

Two mysteries to solve



The sleepy student

Mika skipped breakfast. By 10 a.m., her stomach growls, her head feels heavy, and she can’t focus on the quiz.



The pale pechay

A pot of pechay was left in a dark storeroom for a week. Its leaves turned yellow, and its stems grew weak and droopy.

What do Mika and the pechay both lack?

LEARNING GOALS

Your mission today

1. Explain how **plants make their own food** and how **animals get food** by eating and digesting it.
2. **Trace how nutrients travel** from food all the way into a cell.
3. Describe how cells **release energy from glucose** through cellular respiration.

Key words: photosynthesis · digestion · absorption · diffusion · active transport · mitochondria · cellular respiration · ATP

THE BIG IDEA

Every cell needs two things

Building blocks

Materials to grow, repair damage, and make new cells, like bricks for building a house.

Energy

Power to move, grow, think, and stay alive, like fuel for a jeepney.



ATP: the cell's energy "load"

Cells can't spend food directly. They first turn it into **ATP** (adenosine triphosphate). Think of ATP as prepaid load: no load, no calls!

PART 1 · GETTING FOOD

Two ways to get food



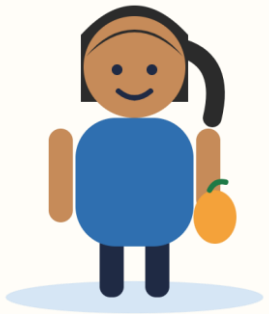
Autotrophs

“SELF-FEEDERS”

Make their own food using sunlight, water, and carbon dioxide.

Examples: mango trees, rice plants, algae

*Meet **Manggo**, the mango tree in our schoolyard.*



Heterotrophs

“OTHER-FEEDERS”

Get food by eating plants, animals, or both.

Examples: humans, carabaos, chickens, fish

*Meet **Mika**, a student who loves rice and ripe mangoes.*

PART 1 · PLANTS · PHOTOSYNTHESIS

Manggo cooks with sunlight

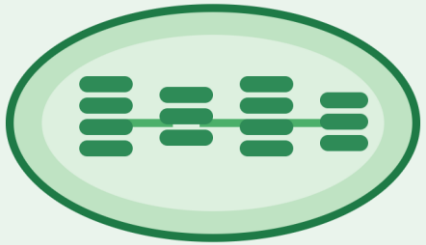
Story: Every sunny morning, Manggo's leaves open tiny pores called **stomata** to take in carbon dioxide, while its roots sip water from the soil.

INGREDIENTS

Light energy
from the sun

Carbon dioxide
from the air, through stomata

Water
from the soil, through roots



The chloroplast kitchen

Green chlorophyll traps light energy.

FINISHED DISHES

Glucose
a sugar: the plant's food

Oxygen
released into the air

$$6CO_2 + 6H_2O + \text{light energy} \rightarrow C_6H_{12}O_6 + 6O_2$$

PART 1 · PLANTS · USING FOOD

Where does Manggo's sugar go?

Burned for energy

Every living cell, from root tip to leaf, breaks down glucose to make ATP, day and night.

Stored as starch

Extra sugar is packed as starch in roots, tubers, and seeds, like camote, cassava, and rice grains.

Built into its body

Glucose becomes cellulose for cell walls and wood. With minerals from the soil, it helps make proteins.

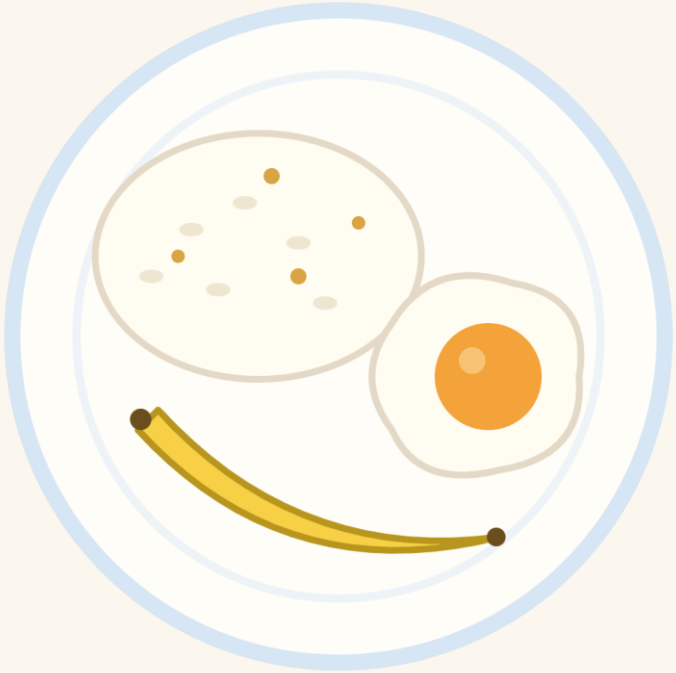
Sweetened into fruit

Sugar sent to the fruit makes a ripe mango sweet. That's why Mika can't resist it!

Delivery routes: sugar travels through the **phloem**; water and minerals travel up from the roots through the **xylem**.

PART 2 · ANIMALS

Mika's breakfast: big molecules



Story: It's 6:30 a.m. Mika sits down to garlic rice, a fried egg, and a banana. Yum!

Garlic rice	→	Carbohydrates (starch)
Fried egg	→	Proteins (and some fat)
Cooking oil	→	Fats
Banana	→	Sugars, vitamins, and minerals

⚠ **The problem:** starch, proteins, and fats are giant molecules, far too big to enter Mika's cells. They must be broken down first.

PART 2 · ANIMALS · DIGESTION

Digestion: food cut down to size

STOP 1

Mouth

Teeth chew food into small pieces. Saliva starts breaking starch into sugars.

STOP 2

Stomach

Strong muscles churn the food. Acid and enzymes start breaking down proteins.

STOP 3

Small intestine

Enzymes finish the job. Bile from the liver breaks fat into tiny droplets.

Carbohydrates → glucose

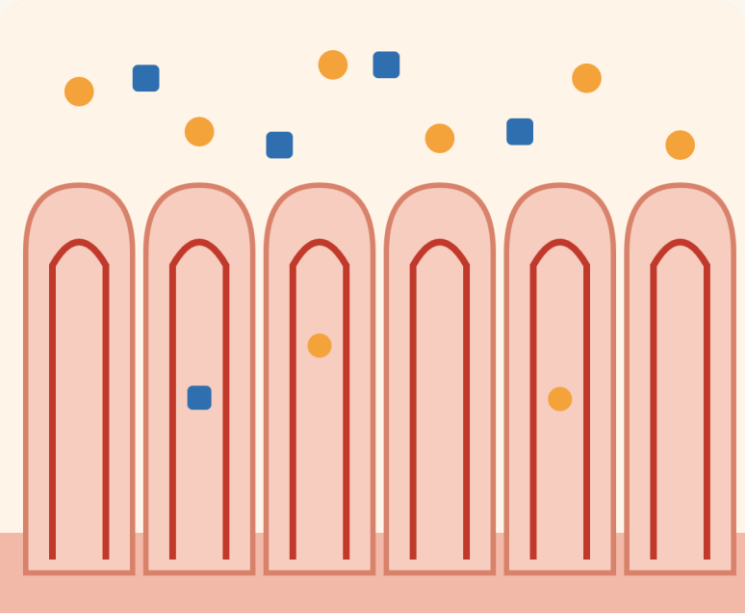
Proteins → amino acids

Fats → fatty acids and glycerol

Think of it like LEGO: the body takes the castle apart into single bricks small enough to fit through a cell's door.

PART 2 · ANIMALS · ABSORPTION

Absorption: into the blood



Villi up close · orange = glucose · blue = amino acids

- The small intestine is lined with millions of tiny, finger-like **villi**.
- Villi create a huge surface area, so nutrients are absorbed quickly.
- Glucose and amino acids pass into tiny blood vessels called **capillaries**. Most fats enter the lymph first.

Blood: the delivery rider

The heart is the hub. Every heartbeat sends glucose, amino acids, and oxygen from the lungs to all of Mika’s cells, from brain to toes.

PART 2 · INTO THE CELL

Getting through the cell's door

Every cell is wrapped in a **cell membrane**, a choosy gatekeeper that lets some things in and keeps others out.



Diffusion

Moves from more to less. No energy needed.

Example: oxygen slipping into Mika's muscle cells



Facilitated diffusion

Uses protein doors. Still no energy needed.

Example: glucose entering Mika's cells



Active transport

Spends ATP to pump things from less to more.

Example: Manggo's root hairs pulling in minerals

PART 3 · PLOT TWIST

Glucose is a locked treasure chest.

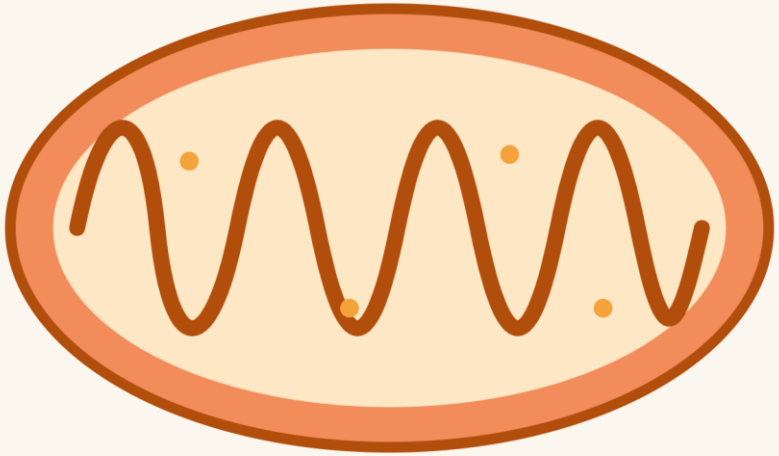
Its energy is stored in chemical bonds. To spend it, cells must unlock it, one step at a time.

The key: cellular respiration



PART 3 · ENERGY · CELLULAR RESPIRATION

The powerhouse: mitochondria



Folded inner membrane = more space to make ATP

Story: As Mika walks to school, thousands of mitochondria in her leg muscles power up.

Glucose + oxygen → carbon dioxide + water + ATP

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + ATP$$

- Happens in **both plant and animal cells**.
- Runs day and night, as long as the cell is alive.
- Some energy escapes as heat. That's why Mika feels warm after running!

PART 3 · ENERGY · INSIDE THE CELL

Three stages, one assembly line

1 · Glycolysis

WHERE: cytoplasm

Glucose splits into two smaller molecules called pyruvate. No oxygen needed yet.

+2 ATP

2 · Krebs cycle

WHERE: inner space (matrix) of the mitochondria

Pyruvate is broken down further. Carbon dioxide is released.

+2 ATP

3 · Electron transport chain

WHERE: folded inner membrane of the mitochondria

Oxygen collects used electrons and forms water.

+32 to 34 ATP

≈ 36–38 ATP from one glucose molecule. It's like breaking a ₱1,000 bill into coins the cell can actually spend!

PART 3 · ENERGY · WITHOUT OXYGEN

No oxygen? Plan B: fermentation

Mika’s 100-meter sprint

Mika runs so fast that her blood can’t deliver oxygen quickly enough. Her muscle cells switch to **lactic acid fermentation**: a little ATP, plus acid that adds to that tired, burning feeling.

Glucose → lactic acid + 2 ATP

The rising pandesal

Yeast in the dough breaks down sugar without oxygen. This **alcoholic fermentation** releases carbon dioxide bubbles that make the bread puff up.

Glucose → alcohol + carbon dioxide + 2 ATP



About 36 ATP with oxygen



Only 2 ATP without oxygen

PART 3 · ENERGY · SPENDING ATP

What does ATP pay for?

In Mika (animals)

- Muscles contract for her jump shot.
- Nerves send signals as she thinks.
- Her heart beats nonstop, day and night.
- New cells heal a scraped knee.

In Manggo (plants)

- Roots and shoots grow longer.
- Root hairs pump in minerals.
- Flowers bloom and mangoes form.
- Damaged leaves are repaired.

Remember: when ATP runs out, cells stop working. That's why food and oxygen keep both Mika and Manggo alive.

PART 4 · THE BIG PICTURE

Partners in the cycle of life

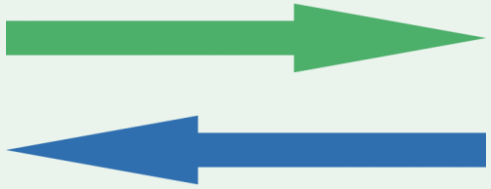


Photosynthesis

In chloroplasts, only in light. Done by plants.

Light energy goes in

Glucose + oxygen



Carbon dioxide + water



Cellular respiration

In mitochondria, day and night. Done by plants and animals.

ATP energy comes out

The **products** of one process are the **raw materials** of the other. Mika breathes out what Manggo needs, and Manggo gives off what Mika breathes in.

PART 4 · COMPARE

Plants vs. animals, side by side

Feature	Plants (Manggo)	Animals (Mika)
How they get food	Make it by photosynthesis	Eat it, then digest it
Where food is made	Chloroplasts, mainly in leaves	Not made; taken in by eating
How nutrients travel	Xylem and phloem	Blood vessels
Where energy is released	Mitochondria (same in both!)	Mitochondria (same in both!)
How extra food is stored	Starch	Glycogen and fat
Plan B without oxygen	Alcoholic fermentation (flooded roots)	Lactic acid fermentation (tired muscles)

PART 4 · THINK AGAIN

Myth busters

MYTH Plants photosynthesize, so they don’t need to respire.	FACT Plants respire all day and night. They photosynthesize only in light.
MYTH The food we eat is already energy our cells can use.	FACT Cells must break food down to glucose, then turn it into ATP.
MYTH Most digestion happens in the stomach.	FACT Most digestion and almost all absorption happen in the small intestine.

PART 4 · CHECK YOUR UNDERSTANDING

Quick check: try it!

1. Which organelle releases energy from glucose in **both** plant and animal cells?

2. Why must Mika’s fried egg be digested before her cells can use it?

3. Manggo’s root hairs pull in minerals using ATP. What is this process called?

4. After a fast sprint, why do Mika’s legs feel tired and burning?

Bonus: The energy in Mika’s rice first came from where?

SUMMARY

The whole journey

Mika (animals)

Eat → Digest → Absorb → Blood → Cell door → Mitochondria → ATP!

Manggo (plants)

Sunlight → Chloroplast → Glucose → Phloem → Cell door → Mitochondria → ATP!

No food, no ATP. No ATP, no life.

TAKE-HOME TASK

Your turn: trace your lunch!

1. Pick one food from your lunch today.
2. Draw a six-panel comic that follows it from your plate to the ATP in your muscles.
3. Add a first panel showing the plant that captured the sun's energy.

Think deeper

If the sun stopped shining, what would happen to Mika's energy? Explain using today's lesson.

Great job, energy experts!